Theoretical Analysis of Straightened Latent Dynamics on Gradient-Based Trajectory Optimization

Technical Report

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Abstract

This document derives the theoretical properties connecting the geometry of latent space trajectories to the convergence rate of gradient-descent based planners in Model Predictive Control (MPC). By bounding the curvature of the latent trajectory tensor (ϵ-straightness), we establish a formal threshold for guaranteed convexification of the planning objective. We analyze the condition number of the optimization landscape under three regimes: (1) a stable $\mathcal{O}(1)$ upper bound for perfectly straight trajectories, (2) a generalized upper bound for approximately straight trajectories ($\epsilon > 0$), and (3) an exponential lower bound $\Omega(e^{cT})$ derived via the Oseledets Multiplicative Ergodic Theorem for unregularized curved trajectories. Finally, we provide empirical methods to estimate these bounds using Hessian-Vector Products (HVPs) and cosine similarity.

1 Problem Formulation

Consider a latent Model Predictive Control (MPC) setup where we optimize a sequence of actions $\mathbf{a} = \{a_0, a_1, \ldots, a_{T-1}\} \in \mathbb{R}^{T \times d_a}$ to minimize a cost over a planning horizon T. Let $z_t \in \mathbb{R}^{d_z}$ denote the latent state at time t. The dynamics are governed by a learned transition function:

$$z_{t+1} = f(z_t, a_t) \tag{1}$$

We define the trajectory mapping $\Phi : \mathbb{R}^{T \times d_a} \to \mathbb{R}^{d_z}$ as the unrolled dynamics mapping the action sequence to the final state $z_T = \Phi(\mathbf{a})$.

The optimization problem is to minimize the final state cost (for simplicity, intermediate costs can be similarly bounded):

$$\min_{\mathbf{a}} J(\mathbf{a}) = C(\Phi(\mathbf{a})) \tag{2}$$

2 Formal Assumptions and Definitions

To analyze the optimization landscape, we make standard assumptions regarding the cost function $C(z)$.

Assumption 1 (Cost Convexity and Smoothness)**.** *The cost function $C : \mathbb{R}^{d_z} \to \mathbb{R}$ is μ-strongly convex and L-smooth. Its gradient is bounded by M in the region of interest:*

$$\mu I \preceq \nabla_z^2 C(z) \preceq LI \tag{3}$$

$$\|\nabla_z C(z)\|_2 \leq M \tag{4}$$

We formalize the geometric concept of a "straight" latent trajectory by bounding its second derivative with respect to the action sequence.

Definition 1 (ϵ-Straightness)**.** *A latent trajectory mapping $\Phi(\mathbf{a})$ is ϵ-straight if the spectral norm of its spatial Hessian for every latent dimension $i \in \{1, \ldots, d_z\}$ is bounded by ϵ:*

$$\|\nabla^2_{\mathbf{a}} \Phi_i(\mathbf{a})\|_2 \leq \epsilon \quad \forall i \tag{5}$$

If $\epsilon = 0$, the trajectory is perfectly straight (locally affine with respect to actions).

3 Convexification Threshold

The convergence of gradient descent on $J(\mathbf{a})$ is governed by the condition number κ of its Hessian $H = \nabla^2_{\mathbf{a}} J(\mathbf{a})$. By the chain rule, H decomposes into a first-order (Jacobian) term and a second-order (Curvature) term:

$$H = \underbrace{J_\Phi^T \nabla^2_z C(\Phi) J_\Phi}_{\mathcal{T}_1} + \underbrace{\sum_{i=1}^{d_z} \frac{\partial C}{\partial z_i} \nabla^2_{\mathbf{a}} \Phi_i(\mathbf{a})}_{\mathcal{T}_2} \tag{6}$$

where $J_\Phi \in \mathbb{R}^{d_z \times (T \cdot d_a)}$ is the Jacobian matrix $\frac{\partial z_T}{\partial \mathbf{a}}$.

Lemma 1 (Convexification via Straightness)**.** *Under Assumption 1, the objective $J(\mathbf{a})$ is strictly convex at $\mathbf{a}$ if the trajectory is ϵ-straight with:*

$$\epsilon < \frac{\mu \sigma^2_{\min}(J_\Phi)}{M\sqrt{d_z}} \tag{7}$$

Proof. By Assumption 1, $\lambda_{\min}(\mathcal{T}_1) \geq \mu \sigma^2_{\min}(J_\Phi)$. The spectral norm of the curvature term $\mathcal{T}_2$ is bounded using the triangle inequality:

$$\|\mathcal{T}_2\|_2 \leq \sum_{i=1}^{d_z} \left| \frac{\partial C}{\partial z_i} \right| \|\nabla^2_{\mathbf{a}} \Phi_i\|_2 \leq \epsilon \sum_{i=1}^{d_z} \left| \frac{\partial C}{\partial z_i} \right| \leq \epsilon \sqrt{d_z} M \tag{8}$$

By Weyl's inequality, $\lambda_{\min}(H) \geq \lambda_{\min}(\mathcal{T}_1) - \|\mathcal{T}_2\|_2 \geq \mu \sigma^2_{\min}(J_\Phi) - \epsilon\sqrt{d_z} M$. Strict convexity requires $\lambda_{\min}(H) > 0$, yielding the stated threshold. □

4 Condition Number Bounds

The condition number $\kappa(H) = \frac{\lambda_{\max}(H)}{\lambda_{\min}(H)}$ dictates the convergence rate of the planner. We establish bounds across three distinct trajectory regimes.

4.1 Ideal Case: Stable $\mathcal{O}(1)$ Bound for $\epsilon = 0$

Theorem 1 (Constant Condition Number for Straight Trajectories)**.** *Assume a perfectly straight latent trajectory ($\epsilon = 0$) where state transitions act as an identity mapping on velocity ($J_z^{(k)} = \frac{\partial z_{k+1}}{\partial z_k} = I$). The condition number of the planning Hessian $\kappa(H)$ is bounded by a constant $\mathcal{O}(1)$ with respect to the planning horizon T.*

Proof. With $\epsilon = 0$, $H = J_\Phi^T \nabla^2_z C(\Phi) J_\Phi$. By the chain rule, the trajectory Jacobian block for time t is $\frac{\partial z_T}{\partial a_t} = (\prod_{k=t+1}^{T-1} I) J_a^{(t)} = J_a^{(t)}$. The temporal product collapses, making the singular values of J_Φ strictly dependent on the single-step action mapping, independent of T. Thus, $\kappa(H) \leq \frac{L}{\mu} \kappa(J_\Phi)^2 \sim \mathcal{O}(1)$. □

4.2 Relaxed Case: Generalized Upper Bound for $\epsilon > 0$

In practice, a neural network cannot achieve perfectly affine dynamics. We relax the assumption to an ϵ-straight trajectory where the temporal Jacobians are approximately the identity.

Assumption 2 (δ-Approximate Identity)**.** *Assume the latent dynamics are trained such that the state-transition Jacobian deviates from the Identity matrix by at most $\delta \ll 1$ in spectral norm: $\|J_z^{(k)} - I\|_2 \leq \delta$.*

Theorem 2 (Generalized Condition Number Bound)**.** *For an ϵ-straight trajectory with a δ-approximate identity transition, assuming the convexification threshold (Lemma 1) holds, the condition number is bounded by:*

$$\kappa(H) \leq \frac{L(\sigma_a^{\max})^2 e^{2\delta T} + \epsilon M\sqrt{d_z}}{\mu(\sigma_a^{\min})^2 e^{-2\delta T} - \epsilon M\sqrt{d_z}} \tag{9}$$

where σ_a are the bounded singular values of the localized action-to-state mapping $J_a^{(t)}$.

Proof. Under Assumption 2, the temporal product of Jacobians over T steps scales as: $\sigma_{\max}(J_\Phi) \leq \sigma_a^{\max} e^{\delta T}$ and $\sigma_{\min}(J_\Phi) \geq \sigma_a^{\min} e^{-\delta T}$. Using Weyl's inequality on $H = \mathcal{T}_1 + \mathcal{T}_2$:

$$\lambda_{\max}(H) \leq \lambda_{\max}(\mathcal{T}_1) + \|\mathcal{T}_2\|_2 \leq L\sigma_{\max}^2(J_\Phi) + \epsilon M\sqrt{d_z} \tag{10}$$

$$\lambda_{\min}(H) \geq \lambda_{\min}(\mathcal{T}_1) - \|\mathcal{T}_2\|_2 \geq \mu\sigma_{\min}^2(J_\Phi) - \epsilon M\sqrt{d_z} \tag{11}$$

Substituting the bounds for the singular values yields the theorem. If δ and ϵ are not heavily regularized, the bound diverges. □

Remark 1 (Dimensionality Scaling and the Curse of Dimensionality).
A critical consequence of the bounds derived in Lemma 1 and Theorem 2 is the presence of the $\sqrt{d_z}$ scaling factor. The spectral norm of the curvature perturbation is bounded by $\|\mathcal{T}_2\|_2 \leq \epsilon M\sqrt{d_z}$, which arises from applying the Cauchy-Schwarz inequality across the sum of all d_z latent dimensions.

Geometrically, this implies a fundamental "curse of dimensionality" for latent trajectory planning. As the representation capacity of the model scales up (larger d_z), even a infinitesimally small uniform curvature ϵ in individual dimensions accumulates into a massive aggregate distortion of the global optimization landscape. Consequently, scaling up high-dimensional unregularized models inherently worsens the condition number and threatens the convexity of the planning objective.

Crucially, enforcing trajectory straightness ($\epsilon \to 0$) completely nullifies this $\sqrt{d_z}$ penalty. By driving the term $\epsilon M\sqrt{d_z} \to 0$, the condition number bound becomes theoretically decoupled from the latent dimensionality. This mathematical property guarantees that straightened latent dynamics can be safely scaled to high-capacity vector spaces without systematically destroying the convergence rate of the gradient-descent planner.

It should be noted that the $\sqrt{d_z}$ factor represents a worst-case isotropic curvature bound (assuming the trajectory bends maximally in all directions aligned with the cost gradient). If the trajectory's curvature is confined to a lower-dimensional principal subspace of effective dimension $k \ll d_z$, the actual empirical degradation of the condition number scales proportionally with $\sqrt{k}$.

4.3 Unregularized Case: Exponential Lower Bound

To prove that the condition number strictly diverges in unregularized non-linear dynamics ($\epsilon > 0, \delta > 0$), we employ Dynamical Systems Theory. Let $J_z^{(k)}$ be generated by a non-linear dynamical system.

Theorem 3 (Exponential Lower Bound on Condition Number)**.** *Let the latent dynamics $z_{t+1} = f(z_t, a_t)$ exhibit standard non-linear dissipative and chaotic behaviors, characterized by a maximum Lyapunov exponent $\lambda_1 > 0$ and a minimum Lyapunov exponent $\lambda_d < 0$. As $T \to \infty$, the condition number of the planning Hessian H is bounded below by an exponentially growing function:*

$$\kappa(H) \geq \Omega\left(e^{2(\lambda_1 - \lambda_d)T}\right) \tag{12}$$

Proof. By the Oseledets Multiplicative Ergodic Theorem, the singular values of the temporal product of Jacobians $J_{z\to z} = \prod_{k=0}^{T-1} J_z^{(k)}$ scale asymptotically with the Lyapunov exponents:

$$\sigma_{\max}(J_{z\to z}) \geq C_1 e^{\lambda_1 T} \tag{13}$$

$$\sigma_{\min}(J_{z\to z}) \leq C_2 e^{\lambda_d T} \tag{14}$$

Assuming the action Jacobian J_a is well-conditioned, the trajectory Jacobian J_Φ inherits this asymptotic scaling. Using Weyl's inequality, we bound the actual condition number:

$$\kappa(H) \geq \frac{\mu\sigma_{\max}^2(J_\Phi) - \|\mathcal{T}_2\|_2}{L\sigma_{\min}^2(J_\Phi) + \|\mathcal{T}_2\|_2} \geq \frac{\mu C_1^2 e^{2\lambda_1 T} - \|\mathcal{T}_2\|_2}{L C_2^2 e^{2\lambda_d T} + \|\mathcal{T}_2\|_2} \sim \Omega\left(e^{2(\lambda_1 - \lambda_d)T}\right) \tag{15}$$

Since $\lambda_1 > 0$ and $\lambda_d < 0$, $(\lambda_1 - \lambda_d)$ is strictly positive, proving the exponential divergence of the condition number. This geometric limitation can only be resolved by driving the Lyapunov exponents to zero via trajectory straightening. □

5 Empirical Estimation Methods

5.1 Exact Estimation via Hessian-Vector Products (HVPs)

Computing the full tensor $\nabla_{\mathbf{a}}^2 \Phi(\mathbf{a})$ is prohibitively expensive. We can efficiently estimate the spectral norm ϵ using Power Iteration with automatic differentiation. For a scalar dimension Φ_i, the HVP with an arbitrary vector v is computed in $\mathcal{O}(1)$ backpropagations:

$$\mathrm{HVP}(\Phi_i, v) = \nabla_{\mathbf{a}}\left(\nabla_{\mathbf{a}}\Phi_i(\mathbf{a})^T v\right) \tag{16}$$

By iterating $u^{(k)} = \mathrm{HVP}(\Phi_i, v^{(k-1)})$ and normalizing $v^{(k)} = u^{(k)}/\|u^{(k)}\|_2$, the associated eigenvalue estimate $\lambda^{(k)} = (v^{(k-1)})^T u^{(k)}$ converges to $\|\nabla_{\mathbf{a}}^2 \Phi_i(\mathbf{a})\|_2$, allowing direct measurement of ϵ.

5.2 Cosine Similarity as a Practical Curvature Proxy

While HVPs evaluate the unrolled sequence, a computationally cheaper proxy during training is the local geometric curvature. Let $v_t = z_{t+1} - z_t$. The empirical cosine similarity is $S = \cos(\theta_t) = \frac{v_t \cdot v_{t+1}}{\|v_t\|_2 \|v_{t+1}\|_2}$.

Assuming bounded velocity norms $\|v_t\| \approx c$, the trajectory spatial curvature is:

$$\|\Delta v_t\|_2^2 = \|v_{t+1} - v_t\|_2^2 \approx 2c^2(1 - S) \tag{17}$$

By Faà di Bruno's formula, the action-Hessian spectral norm ϵ is bounded by the compounding spatial curvature over T steps plus the intrinsic policy smoothness L_a:

$$\epsilon \leq \mathcal{O}\left(T \cdot c\sqrt{1 - S}\right) + L_a \tag{18}$$

Thus, maximizing empirical similarity $S \to 1$ directly suppresses the trajectory spatial curvature $\sqrt{1-S} \to 0$. This acts as a proxy for minimizing ϵ, enforcing the conditions necessary for Lemma 1 and avoiding the exponential divergence modeled in Theorem 3.